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Composite film with hollow hierarchical silica/perfluoropolyether filler and surface etching for performance enhanced triboelectric nanogenerators

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ABSTRACT

Triboelectric nanogenerators (TENGs), capable of harvesting distributed energy from the environment, have provided a viable solution for self-powered wearable electronics. However, their low power density and susceptibility to contamination still severely limit their practical application. Herein, we reported a "two-step" strategy to implement a waterproof self-powered wearable electronic device based on TENG. Firstly, Perfluoropolyether (PF) and hollow hierarchical silica (H-SiO_2) as fillers significantly improve the dielectric properties of polydimethylsiloxane (PDMS) composite films (PFS films) through polarization effect. Then the nano bumps of 10–30 nm are created on the surface of the films by surface etching, the bumps can increase the contact area of TENG. The voltage of TENG is improved from ~ 67 V to ~ 910 V and the current is improved from ~ 1 μA to ~ 25 μA. What's more, the investigation of the surface structure of the PFS composite films by Atomic Force Microscope (AFM) revealed that the synergistic combination of multiple nanoscales (10–30 nm, 60–80 nm, 300–500 nm) is more favorable to achieve high performance of TENG. The surface structure composed of multiple nanoscales endows PFS with excellent superhydrophobicity. The PFS films remained fine performance after 40,000 drops of water and 90 days. This work achieves high performance and water resistance of TENG, demonstrating great potential in the field of self-powered wearable electronic devices.

1. Introduction

Wearable technologies have significantly facilitated various parts of people's life owing to their convenience, including medical monitoring, communication, social media, warning system and interface interaction, etc [1–4]. However, how to sustainably power these wearable electronic devices is a key issue. As an emerging energy harvesting technology, TENGs can convert distributed energy into electrical energy, and has become a preferred choice for the powering miniaturized sensors [5–9]. In particular, the wide selection of materials and diversification of structures have significantly enhanced the applicability of TENGs in the field of wearable devices. However, the output power density of TENGs still needs to be furtherly optimized [10,11]. The low output power remains a significant barrier to commercial implementation of TENGs, it is of positive significance to construct the high performance of TENGs [7].

To date, researchers have developed a variety of methods to improve the output power density of TENGs, including fabrication of surface micro/nanostructures, surface charge injection, and the processing and manufacture of composite materials [12–14]. Filling the polymer with nanomaterials not only enhances the capacity of polymers to acquire or lose electrons, but also changes the surface morphology and increases the contact area of TENGs, this is an effective approach to enhance the output power density of TENGs [15–20]. Due to the rapid advancement of nanomaterial preparation, a variety of nanomaterials have been used as fillers in conjunction with polymers to prepare composite materials as triboelectric layers of TENGs, including carbon nanotubes (CNTs), MoS_2, $BaTiO_3$, graphene, TiO_2, MOF, MXene, hydrogel, and black phosphorus [15–23]. However, these fillers are all single micro or nanoscale, and exploring nanomaterials with hierarchical or multiple scale as fillers to

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improve the performance of TENGs remains a significant challenge.

Due its flexibility, friction resistance, transparency and hydrophobicity, the material of polydimethylsiloxane (PDMS) has been widely employed in the field of flexible electronic devices [24,25]. Nano hollow materials are functional materials with certain voids or intervals inside. Because of their low density, high specific surface area, small space gap, and rapid electron transmission rates, they offer tremendous promise in the domains of energy storage, photoelectric catalysis, and biosensing [26]. It's a hot topic in the manufacturing of PDMS materials to combining PDMS with hollow materials for constructing functionalized biomimetic superhydrophobic surface structures through covalent bonding, hydrogen bonding, and polarization between film molecules and particles [24,27,28]. Especially in recent years, the PDMS-based superhydrophobic TENGs have been constructed by stencil etching and 3D printing, and these devices were used to harvest mechanical energy, raindrop energy and aqueous fluid sensing [9,29,30]. Assuring continuous and stable performance of TENGs in harsh environments is a serious issue. Especially in high humidity, the water layer created at the polymer triboelectric-electric interface may impair charge transmission or result in charge dissipation, which necessitates a hydrophobic triboelectric layer [31,32]. However, due to the fine micro/nano structure of superhydrophobic surface, their ability to withstand the impact of water droplets and long-term remains unsolved.

In this work, a high-performance TENG based on PDMS composite film was proposed. By combining polarization and surface microstructure etching, the electrical output was greatly improved, and the composite film exhibits exceptional superhydrophobicity. During the curing process of PDMS, the abundant hydroxyl groups (–OH) on the surface of H-SiO_2 and -CF_3 in Perfluoropolyether (PF) form lots of hydrogen bonds with the –CH_3 in PDMS side chains, resulting in the free association form of –CH_3, forming the interfacial polarization between SiO_2 and PDMS molecular chains and the molecular polarization between PF and PDMS molecular chains. Attributed to the synergistic effect of the two polarization methods, the relative dielectric constant of PDMS composite film was improved from ~ 3.1 to ~ 11.03. And the surface etching greatly increased the contact area of the TENG. Based on the polarization and etching strategy, the output was improved from ~ 67 V to ~ 913 V. The investigation of the surface structure of the PFS composite films by AFM has revealed that the synergistic combination of multiple nanoscales (10–30 nm, 60–80 nm, 300–500 nm) is more favorable to achieve high performance of TENGs. The TENGs can charge capacitors with different capacities and harvest human kinetic energy. In addition, this composite membrane exhibits excellent superhydrophobicity and stability, and its output remains 90% after 40,000 drops of water impact and 90 days. This work presents an effective method to enhance the output performance of TENG through polarization and surface microstructure etching, and guarantees the waterproofness of the triboelectric layer, showing great potential in the field of waterproof self-powered flexible wearable electronics.

2. Methods & experimental

2.1. Materials

Glucose monohydrate (AR, M_w = 198), PF oil (AR, M_w = 6000), Silicon tetraacetate (TEOS, AR, 95%) were purchased from Shanghai Macklin Biochemical Co., Ltd. PDMS precursors (Sylgard 184, Dow Corning), n-Heptane (AR, 98%), ethanol (AR, 99.9%) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Copper foil (4.5 × 4.5 cm^2) were purchased from Tianjin Gaoke new material technology Co., Ltd. All chemical reagents were used as received.

2.2. Fabrication of H-SiO_2

The 7 g of Glucose monohydrate was added into 48 mL of deionized water and continuously stirred for 30 min o form a homogeneous suspension. The homogeneous suspension was then transferred into a 100 mL Teflon-lined autoclave and processed at 180 °C for 14 h, then cooled to room temperature. The precipitate was washed several times through water and ethanol, and then freeze-dried for 20 h to obtain dried carbon nanospheres(C).

The hierarchically mesoporous silicon dioxide(H-SiO_2) was obtained as follows: 0.1 g of carbon nanospheres was dissolved in the mixture of 80 mL of deionized water and 20 mL of ethanol, and then 2 mL of TEOS was added in the above solution under vigorous stirring, the reaction lasted 12 h, and the precipitate was obtained by washing several times by deionized water and ethanol. The precipitate was product by annealing at 400 °C for 2 h and cooling down to room temperature.

2.3. Fabrication of PFS films

A certain amount of H-SiO_2 is added to the mixture of 20 mL n-heptane and 1 mL ethanol to form a suspension, being stirred continuously for 15 min. 0.05 g of PF and the suspension were added to 10 mL of PDMS base elastomer, the mixture was heated with 100 °C for 2 h with stirring to remove n-heptane, then cooled to room temperature. The curing agent was added to the mixture for thorough stirring. Then the mixture was spin-coated on the copper foil at 3000 rpm for 13 s, finally cured on the copper foil at 80 °C for 4 h to obtain the PFS composite films.

2.4. Construction of TENGs

The composite film cured on the copper foil serves as one triboelectric layer of the TENG, the copper foil serves as the electrode layer, and the other triboelectric and electrode layer is selected directly from the same copper foil, finally form a conventional contact-separation TENG.

2.5. Working mechanism of TENG and the influence of dielectric constant on its output

The working mechanism of TENG adopts the capacitor structure, and according to previous reports, the output voltage of TENG can be expressed as the following equation:

$$V = \frac{(\sigma_0 - \Delta\sigma)\bullet x(t)}{\varepsilon_0} - \frac{\Delta\sigma\bullet d_{film}}{\varepsilon_r\sigma_0}$$

ε_0, ε_r, σ_0 and Δ_σ are the vacuum permittivity, the relative permittivity of the film, the surface charge density of of triboelectric layer on the film and the transfer charge density on the electrode, respectively. Since there is no charge transfer at the open-circuit voltage, $\Delta\sigma$ is 0 and the equation of open-circuit voltage of TENG can be expressed as:

$$V = \frac{\sigma_0\bullet x(t)}{\varepsilon_0}$$

and because the TENG can act as both an output and an energy storage device, σ_0 depends on the capacitive properties of the TENG, whose capacitance is determined by the following equation:

$$C = \varepsilon_0 S\bullet\frac{\varepsilon_r}{d_{film}}$$

The relative permittivity of the film ε_r is proportional to the capacitance C, which indicates that the open-circuit voltage of TENG is also proportional to ε_r. From this, the permittivity of the triboelectric layer can change the electric output of TENGs, the electric output of TENGs can be enhanced by increasing the dielectric constant.

2.6. Characterization of the materials and TENGs

The surface morphology and Energy Dispersive Spectroscopy (EDS) of PDMS composite films were characterized by Field Emission Scanning

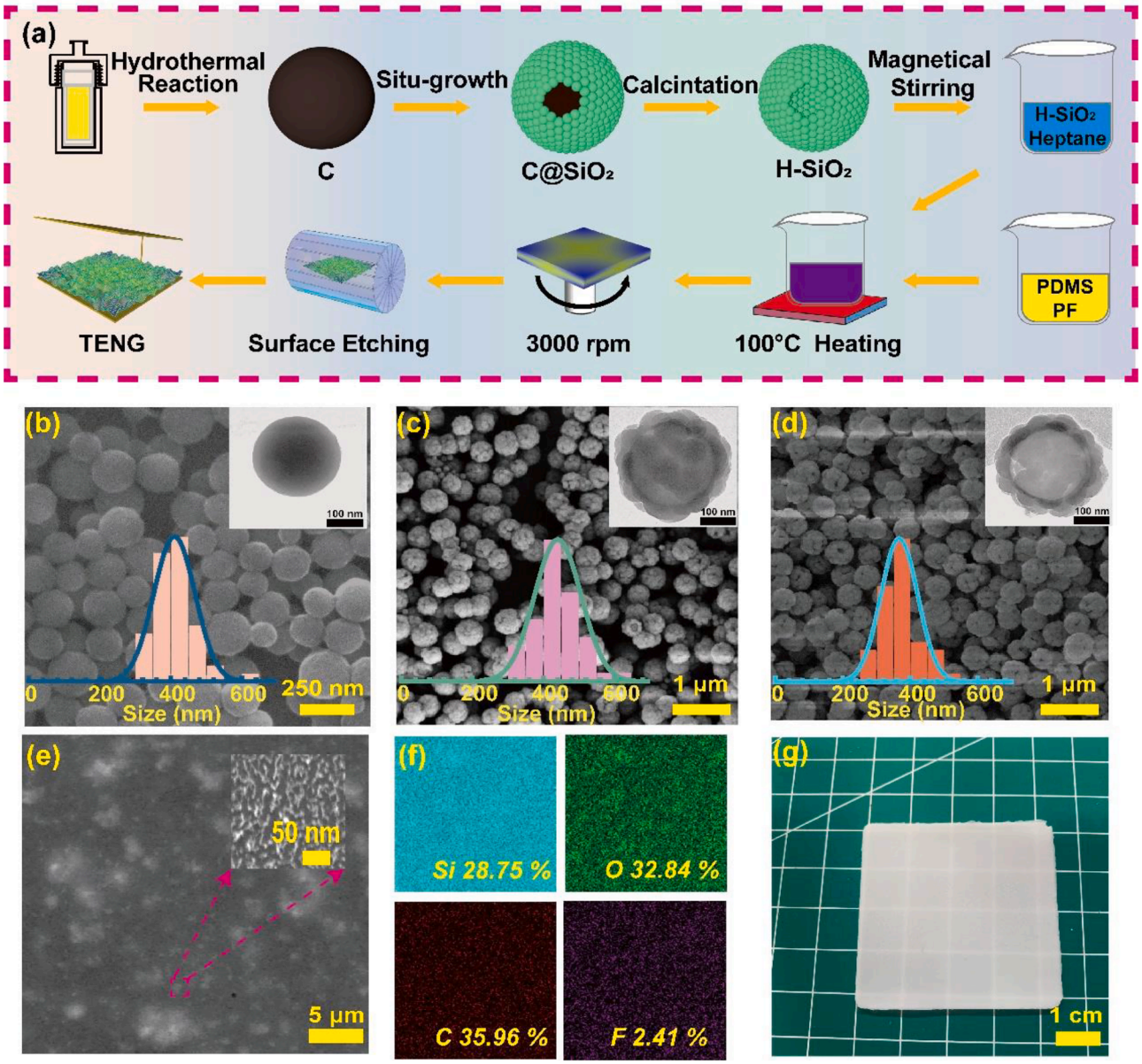


Fig. 1. Construction of PFS films and TENGs. a, Schematic diagram for the construction of H-SiO_2 PFS films and TENGs based on the PFS films. b-d, SEM images and the size distributions of C, C@SiO_2, H-SiO_2 (the inset show the TEM images of C, C@SiO_2, H-SiO_2). e, SEM image of PFS (8%). f, element distribution maps of PFS (8%) films. g, photographs of the PFS (8%) films.

Electron Microscope (SEM, SU8020, Hitachi, Japanese) and transmission electron microscopy (TEM, TecnaiF20, FEI, United States). The dielectric constant was measured through a parallel plate capacitor model by using a Keysight HC-E4990A Semiconductor Device Parameter Analyzer. X-ray diffraction (XRD) spectra were acquired by X Pert Pro (Holland) X-ray diffract meter with Cu Kα1 radiation (λ = 0.154 nm) from 10° to 40°. Infrared spectra were collected with an FTIR spectrophotometer (Bruker Vertex 80 V). N_2 adsorption–desorption was performed on a Micromeritics ASAP 2020 instrument (AUTOSORB IQ, United States). Nano bumps were constructed by the SUNJUNE PLASMA VP-R5 in argon atmosphere. The roughness measurement was carried out using an Asylum Research MFP-3D-SA AFM. Tensile strain tests were carried out by C610M intelligent electronic tensile tester. Electrical performance was measured by a low-noise electrometer (Keithley 6514, USA and 347–3-H-CE, USA). The pressure about 20 kPa applied on the TENGs to provide the contact–separation motion by the linear motor (XRV97-AVA2-40-EC-UDMLC).

3. Results and discussion

3.1. Construction of TENGs based on the PFS films

The PFS films and the TENGs were prepared as shown in Fig. 1a. First, the carbon nanosphere (C) were prepared under hydrothermal conditions. Briefly, low-cost glucose was chosen as the carbon source, after 14 h of hydrothermal reaction and 20 h of freeze-drying, the C was obtained [33]. Both scanning electron microscopy (SEM) image and transmission electron microscopy (TEM) image showed the successful preparation of carbon spheres, which were uniformly dispersed and concentrated in the size range of 300–500 nm (Fig. 1b) and it can be seen from Figure. S1a that hydrothermal carbons are brownish yellow. Subsequently, C@SiO_2 was obtained by in-situ growth using ethyl orthosilicate as the precursor. The Fig. 1c shows that the size of the C@SiO_2 was concentrated in range of 300–600 nm, and TEM image shown that spherical SiO_2 with a size of 60–90 nm covered the surface of C spheres,

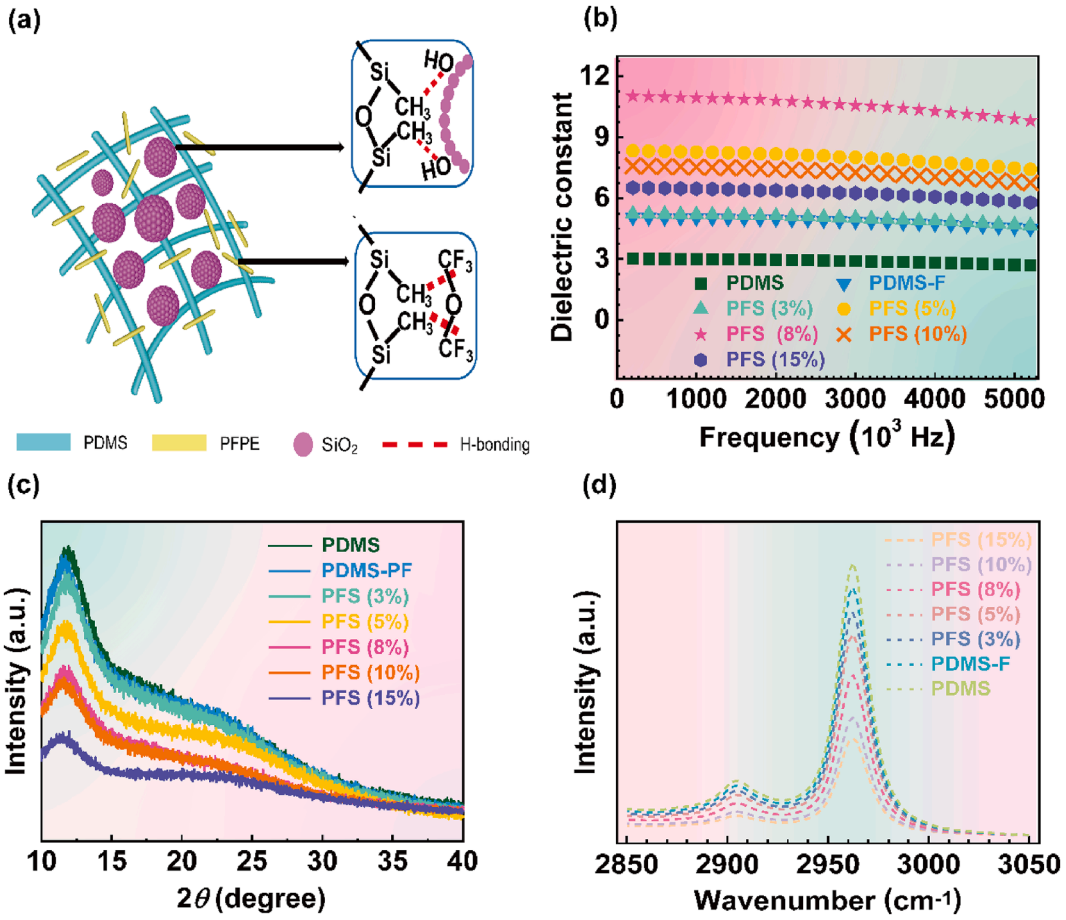


Fig. 2. Mechanisms of polarization. a, Hydrogen bonding between PDMS and H-SiO_2 particles/PF molecule. b, dielectric constant of PDMS, PDMS-F and PFS (3%, 5%, 8%, 10%, 15%) films. c, XRD patterns of PDMS, PDMS-F and PFS (3%, 5%, 8%, 10%, 15%) films. d, integrated area fitting of $-CH_3$ groups in films.

the appearance color of C@SiO_2 composite are also brownish yellow (Figure. S1b). The SiO_2 deposited by in-situ growth formed two nanoscale roughness on the surface of carbon nanospheres, forming a hierarchical structure. After annealing treatment, the hierarchical hollow (H-SiO_2, white powder, Figure. S1c) with an average diameter of about 400 nm was obtained (Fig. 1d).

Fig. 1e and f shows the SEM images and elemental distribution of the PFS (8%) films. The intercalation of H-SiO_2 with PDMS formed some agglomerates. The EDS analysis demonstrates the main elements of PDMS, H-SiO_2 and perfluoropolyether, the existence of F element proves the success of PF inclusion. Fig. 1g shows the photograph of the PFS film, it had transparency and the size was 4.5 cm × 4.5 cm^2, Figure. S2 also shows the thickness of the film PFS(8%), five positions were measured and the thickness were 22.95, 23.97, 23.97, 24 and 24.9 μm respectively.

3.2. *Filler enhances the electrical output of TENGs*

As the triboelectric layers for TENGs, the PDMS with low dielectric constant cannot maintain high charge density on the surface, it greatly restricts the electrical output of TENGs. The intercalation of other materials is an effective way to improve the dielectric constant of PDMS by polarization effect, such as the intercalation of PDMS with graphene, $BaTiO_3$, monolayer MoS_2, titanium dioxide, MOF, etc [14,21,34,35]. Even small amounts of these particles, molecules, or ions can significantly affect the physical and chemical properties of PDMS polymer films. To investigate the effect of fillers on PDMS performance and TENGs applications, we prepared PDMS blends using PF and H-SiO_2 as the fillers. PF is a Krytox oil with excellent frictional properties, thermal stability, mechanical flexibility and hydrophobicity [36]. Due to the large amount of fluorine in the main chain, its ability to obtain electrons is strong. We first examined the dielectric constants of the perfluoropolyether and PDMS blends. The samples of the blends were assigned with the symbol "PDMS-F(X)", where X represents the mass percentage of perfluoropolyether. Fig. 2b and S3 shows the dielectric constants of PDMS-F composite films, with the maximum value at different frequencies coming from a sample with 5% perfluoropolyether, which is attributed to the formation of hydrogen bonds between the H atom in the $-CH_3$ group of the PDMS molecular chain and F atom in the perfluoropolyether (Fig. 2a). The H-SiO_2 has a hierarchical structure with two scales of 60 nm and 300–600 nm (Figure. S4a), its hollow structure is more favorable for electron/proton transport, with a larger specific surface area and a balanced pore size distribution from microporous to mesoporous (Figure. S4b, c). As shown in Fig. 2a, the hydroxyl groups on the surface can form dense hydrogen bonds with H atoms in the PDMS side-chain $-CH_3$ group [14,37]. The dense hydrogen bonding network hinders the directional solidification of PDMS molecular chains and reduces the crystallinity of PDMS (Fig. 2c), the different electron cloud densities in different crystallization intervals are more favorable to the formation of interfacial polarization and molecular polarization, inducing higher dielectric constants [38]. Based on this, the dielectric constant of PDMS was further improved by intercalating different contents of H-SiO_2 at the 5% perfluoropolyether content. The samples of the blends are designated according to the symbol "PFS(X)", where P stands for PDMS, F stands for 5% of perfluoropolyether, and X stands for the mass percentage of H-SiO_2. The dielectric constant of PFS composite membrane is shown in Fig. 2b. The dielectric constant of the PFS composite membrane is higher than that of the pure PDMS membrane, in

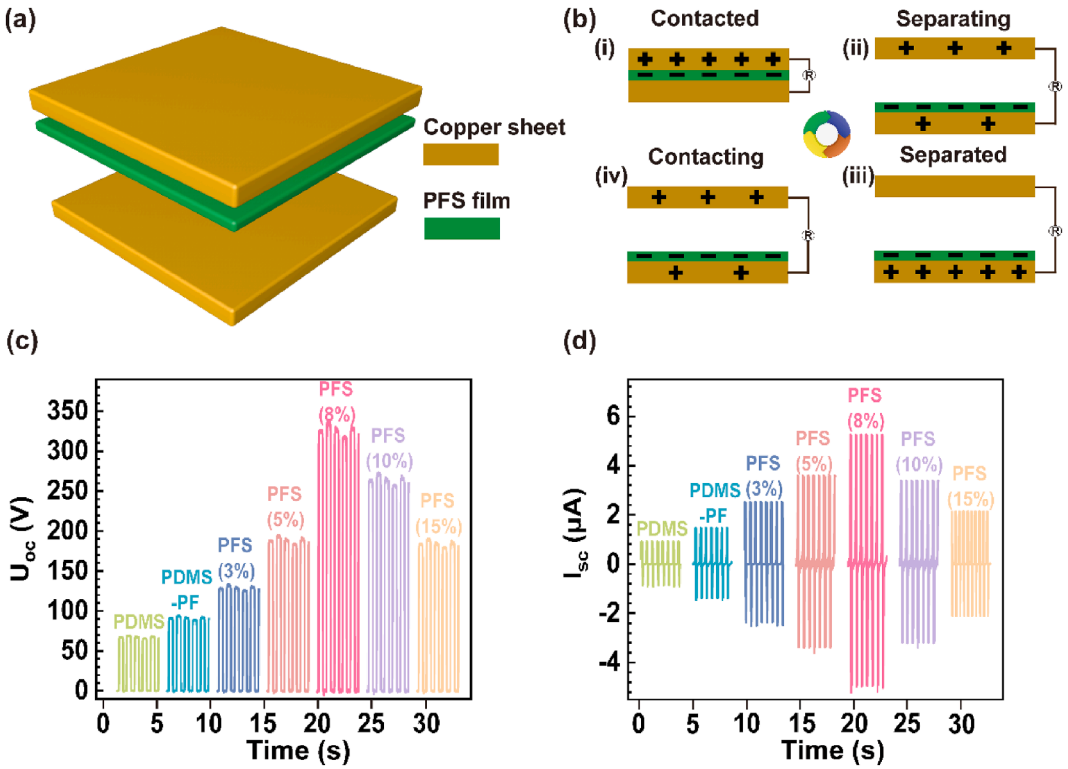


Fig. 3. Working principle of the TENGs in contact-separation mode and enhanced output characteristics with different dielectric constants. a, the 3D schematic of assembled TENGs. b, schematic diagram of the working mechanism of TENGs. c, the open-circuit voltage of TENGs. d, the shot-circuit current of TENGs.

which the relative value of H-SiO_2 with 8% content is about 3.5 times higher than that of the pure PDMS at the same frequency. We performed XRD and FTIR characterization of all the above samples. Fig. 2c shows the degree of crystallinity of the samples, with the highest peak located at 11.5°. The peak intensity of pure PDMS is higher than that of all PDMS composite films, indicating that the higher crystallinity of pure PDMS and the presence of hydrogen bonds prevent the directional curing of PDMS molecular chains. FTIR was used to characterize the relative intensity of $–CH_3$ groups in the membranes, and Gaussian distribution curves were used to fit the area size of $–CH_3$ groups in the membranes, as shown in Fig. 2d. The relative integral area of $–CH_3$ located at 2905 cm^{-1} and 2961 cm^{-1} decreased with the introduction of perfluoropolyether and the increase of H-SiO_2 intercalation content, indicating the formation of dense hydrogen bonding network in the composite membranes. The introduction of inorganic filler H-SiO_2 decreases the mechanical flexibility of PDMS films, but the intercalation of perfluoropolyether can enhance the mechanical flexibility of the films, as can be seen in Figure. S5 and S6. The introduction of perfluoropolyether greatly enhances the mechanical flexibility of PDMS films (2.5 times that of pure PDMS), and the mechanical flexibility of the film with the highest dielectric constant (FPS (8%)) is still higher than that of pure PDMS films, and the mechanical flexibility of FPS films at 15% H-SiO_2 content is close to that of pure PDMS films.

The 3D schematic of assembled TENG is shown in Fig. 3a. The TENG has a typical contact-separation structure with two triboelectric layers of PFS film and copper (the copper foil is also used as electrode, and the electrode at the bottom of PFS film is also copper foil). Copper foil with dimensions of 4.5 × 4.5 cm^2 as the triboelectric layer and electrode, and the PFS film as the triboelectric layer and its substrate copper foil with dimensions of 4.5 × 4.5 cm^2 as the electrode, forming a contact separation TENG with a linear motor. As shown in the Fig. 3b, based on the recognized working mechanism in contact-separation mold of TENG (contact electrification and electrostatic induction):(i) Under the driving of external force, equivalent charges of opposite polarity will be formed on the two contact surfaces of copper and PFS. (ii) and (iii)When the copper and PFS are separated, an air gap will be formed in the middle and a potential difference will be formed between the two copper electrodes, so as to drive electrons from the copper electrode of the substrate to the electrode as the triboelectric layer, forming a reverse potential difference to balance the electrostatic field. (iv) When the air gap between the triboelectric layer copper and PFS is closed, the potential difference formed by the friction charge disappears and electrons reflux occurs [5]. This output performance of all film samples, including open-circuit voltage and short-circuit current of TENGs, were systematically studied as shown in Fig. 3c and d. As a result, this FPS (8%) holds the maximum electrical output, with a sustained output of 320 V open-circuit voltage (U_{oc}) and 5 μA short-circuit current (I_{sc}) with an area of 4.5 × 4.5 cm^2, which is 4.6 times and 5 times higher compared with the pure PDMS (70 V, 1 μA). As shown in the Fig. 3c and d, the electrical output is higher than that of pure PDMS for both purely intercalated fluorine filler and other different contents of H-SiO_2, but as the content of H-SiO_2 reaches 10% and 15%, the output is lower than the PFS (8%), which is due to the excess hydrogen bonding network that restricts the migration of polarized group molecules and thus reduces the dielectric constant of the films, the surface charges of different films were investigated, and their numerical trend are the same as the open-circuit voltage and short-circuit current of TENGs (Figure. S7).

3.3. *Multilevel nanoscale enhances the electrical output of TENGs*

Micro/nano structures on the surface of the triboelectric layer can be fabricated using surface deposition, laser etching, self-assembly, and other technological ways to enhance the contact area and construct high performance of TENGs, it is a promising approach. As shown in Fig. 4a, the PFS (8%) films were used as the carrier, and nano-bumps of 10–30 nm are fabricated on its surface by micro etching technique. These nano-bumps were beneficial to increase the contact area of the triboelectric layers and enhance the output performance of TENG. The output

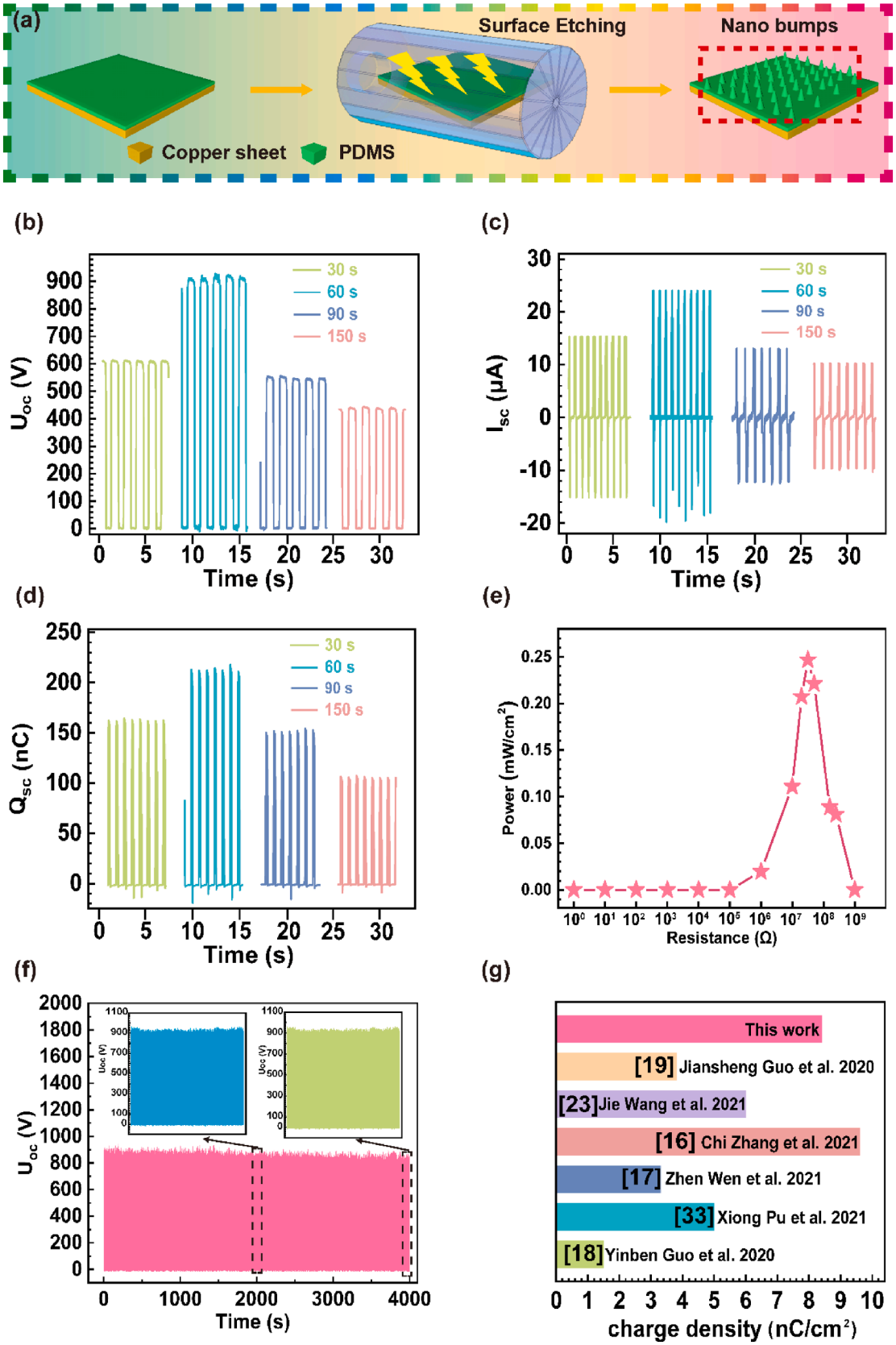


Fig. 4. Output performance of the fabricated TENG with surface etching. a, schematic diagram of surface etching. b-d, the U_{oc}, the I_{sc} and the Q_{sc} of TENGs with surface etching at different times (30 s,60 s, 90 s, 150 s). e, the output power of the TENGs (PFS (8%)) with external resistance. f, the durability test of the TENGs (60 s). g, the comparison of the surface charge density obtained in this work with other reports.

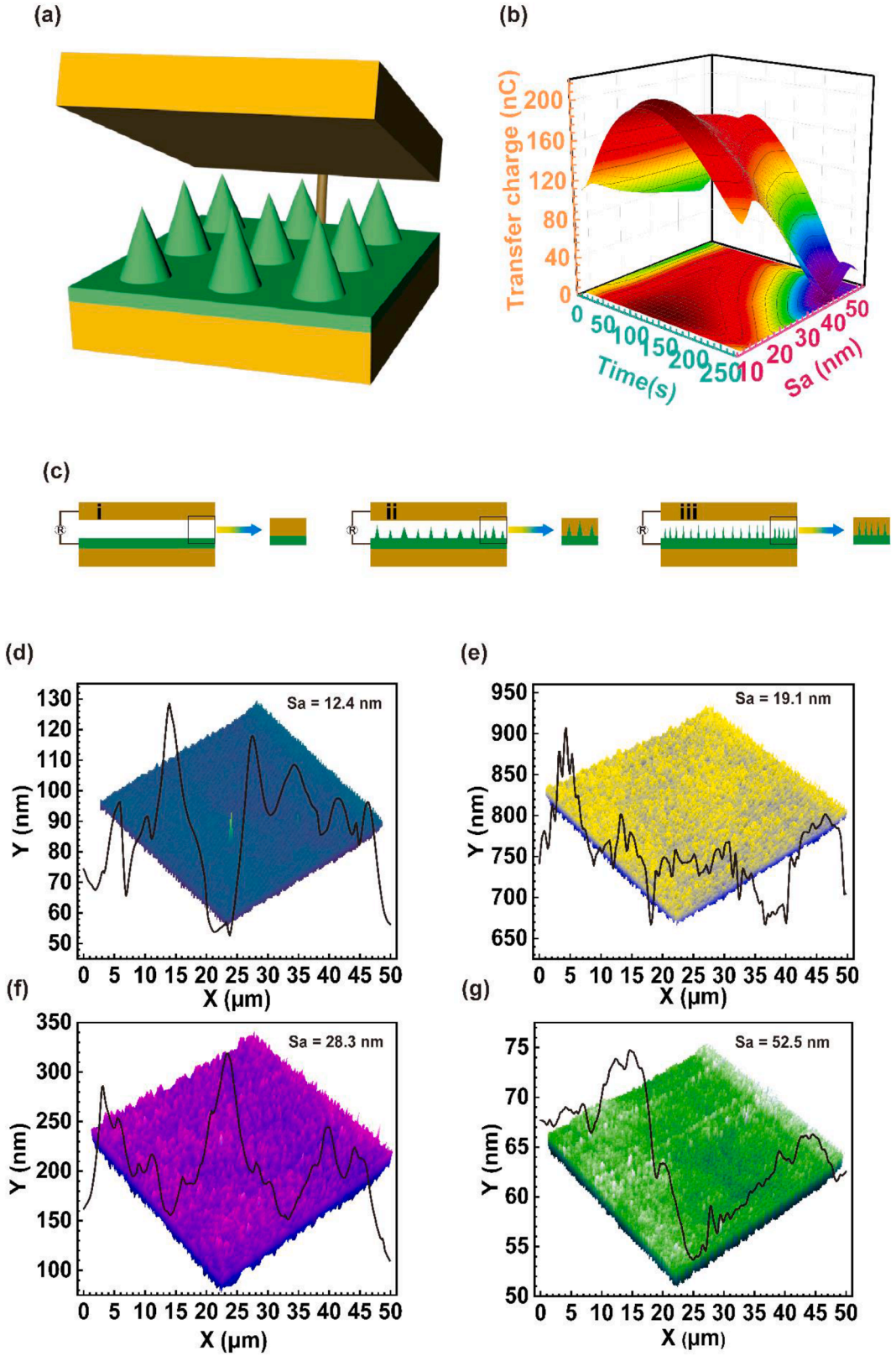


Fig. 5. Synergistic combinations of multiple nanoscales. a and c, schematic diagram of surface microstructure and microstructure contact. b, three-dimensional fitting curve and distribution of transfer charge, etching-time and surface roughness. d-g, roughness of PFS (8%) films with different etching-time.

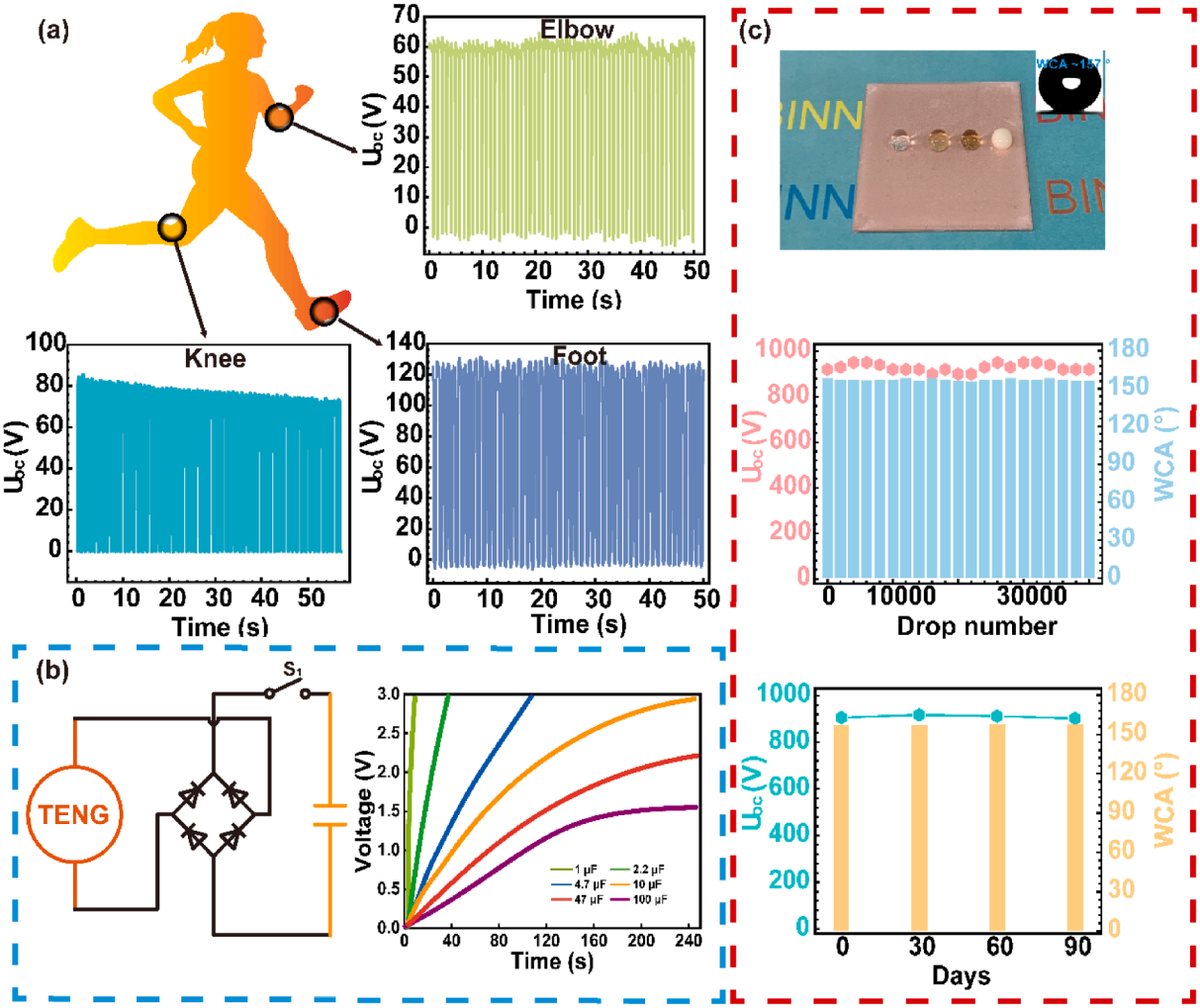


Fig. 6. Applications and super hydrophobicity of PFS(60 s)-TENGs. a, demonstrations of the PFS(60 s)-TENGs were used to harvest biomechanical energy from different parts of the human body (elbow, knee, foot). b, equivalent circuit of a self-powered system using the energy harvested from the TENG for charging capacitors of 1, 2.2, 4.7, 10, 47 and 100 μF. c, photographs of water, juice, cola and milk on the PFS (8%) film, electrical output of TENG and WCA of film after the impact of 40,000 drops of water, electrical output and WCA of PFS film in 0–90 days.

performance of TENGs with different processing times (30 s, 60 s, 90 s, 150 s), including U_{oc}, I_{sc}, and Q_{sc}, were evaluated systematically. As shown in Fig. 4b-d, the TENGs with the processing time of 60 s has the highest performance with continuous output of 910 V, 24 μA, 210 nc, and in the Video. S1 a continuous and huge output of 910 V and 24 μA were shown. Subsequently, the output power of PFS(60 s)-TENGs was measured at 1 to 0.9 GΩ external load, as shown in Fig. 4e and Figure. S8, as presented Figure. S8, it can be told that the voltage and current keep stable when the external load under 1 MΩ. And between 1 and 100 MΩ, both the voltage and current varied sharply and gradually back to stable state after the load more than 100 MΩ. The PFS(60 s)-TENGs had a maximum output power of 0.25 mW/cm^2 at the external load of 28 MΩ, showing great output performance. Additionally, the durability test was carried out, as seen in Fig. 4f. After about 6000 cycles of testing, the U_{oc} demonstrated excellent stability, demonstrating excellent stability of TENG. In order to further prove the stability of surface structure of the PFS(60 s), the surface structure and element composition after 0, 3000 and 6000 cycles were investigated respectively. As shown in Figure. S9 the nano bumps was not seriously damaged, and the content of Si, O, C and F changed only slightly. We compared the surface charge density of this work with recent reports using similar methods as seen in Fig. 4g, the values of this work show certain superiority.

Additionally, as seen in Fig. 4b-d, the output of PFS(60 s)-TENGs were greatly boosted at 30 s, 90 s, and 150 s etching-time. The output obtained with 30 s etching-time is 620 V, 15 μA, with 90 s is 550 V, 13.5 μA, with 150 s is 440 V, 10 μA. These output performances were related to the surface roughness of films, and in this case, AFM was utilized to systematically analyze the surface roughness of films. Compared to the single micron scale of pure PDMS film, the PFS films had a higher roughness (Figure. S10), which was also reflected in the output performance of TENGs made of the thin films (Fig. 3b). For nano-bumps, such as those shown in Fig. 5a and c, are fabricated on PFS flexible films, enhancing the output performance of TENGs. Subsequently, the surface roughness of the PFS films with different etching-times were evaluated as shown in Fig. 5d-g. The surface roughness of the PFS films increased with the etching-times, and roughness were 12.4 nm, 19.1 nm, 28.3 nm, and 52.5 nm, respectively. However, the output performance of TENGs were not increased with the increasing roughness, reaching a maximum at 60 s etching-time, which was attributed to the fact that the multi-level nanoscale is more favorable for full contact with the solid surface. As shown in Fig. 5 ci, when the PDMS surface is pure micron scale, the contact area is restricted; however, when the etching-time is excessive (150 s), as shown in Fig. 5cii and iii, the surface scale becomes more concentrated in nano bumps (10–30 nm), reducing the contact area of solid surface again. As shown in Fig. 5c, when the etching-time are 30 s, 60 s, and 90 s, the surface scale comprises of thin films, particles, and

bumps. In the nanoscale range, roughness values of 10–30 nm, 60–90 nm, and 300–500 nm are more balanced and favorable to full contact with the solid surface at low frequencies, where the output performance of TENGs is optimal with 60 s etching-time. Based on our experimental results, we performed a three-dimensional fit of the relationship between etching-time, surface roughness and transferred charge, as shown in Fig. 5b, more charge transfer occurs when the roughness lies within 15–30 nm.

3.4. Application and superhydrophobicity of PFS(60 s)-TENGs

With high output performance and multiscale surface, the PFS(60 s))-TENGs were resistant to water and has great potential in the field of flexible wearables. This PFS(60 s)-TENGs can be adhered to the elbow, underfoot and knee of an adult to collect human mechanical energy. Demonstration of the voltage generated during the three movements of elbow flexion, knee flexion and walking is shown in Fig. 6a. In Fig. 6b, we employed the PFS(60 s))-TENGs as a power source for various capacitors (1, 2.2, 4.7, 10, 47,100 μF), connected to a rectifier bridge and charging the capacitors. The capacitor charging curve demonstrates that our high performance PFS(60 s))-TENGs has fine practical applicability. In addition, in the Video. S1 the PFS(60 s))-TENGs were used for continuously powering some low-power electronic devices, including calculator, electronic watch, thermometer, indicating the great potential application. The surface of conventional flexible electronic devices is susceptible to contamination by water and other liquids. Due to the multiple scales of, our PFS film is an ideal carrier for superhydrophobic fabrication. We investigated the repellency of PFS (60 s)) to a variety of liquids including water, juice, tea, and cows, as shown in Fig. 6c, where a water contact angle (WCA) of 157° showed excellent superhydrophobicity and repellency to other liquids as well (Table S1). We also have tested the stability of the superhydrophobicity of PFS(60 s) films. As shown in Video. S2 and Fig. 6c, after 40,000 drops of water impacting surface of the PFS films, the open circuit voltage and WCA remained almost unchanged, while the structure and element of PFS(60 s) films had not suffered damage and change(Figure. S11). More importantly, after up to 90 days, the output performance and water resistance of PFS(60 s)-TENGs showed excellent stability. The roll-off angles of water at 0, 30, 60, 90 days were summarized in Table. S2, the changes of roll-off angles were little, indicating the stability of the superhydrophobicity of PFS(8%) films, while the dynamic change of water hitting the PFS surface at 90 days also proved that (Figure. S12).

4. Conclusions

In summary, we proposed a “two-step” strategy to achieve high performance of TENGs and that their triboelectric layers are not damaged by moisture. Firstly, Perfluoropolyether (PF) and hollow hierarchical silica (H-SiO_2) as fillers significantly improve the dielectric properties of polydimethylsiloxane (PDMS) composite films (PFS films) through polarization effect. The dielectric constant of the PFS (8 %) films was increased by 3.5 times compared to the pure PDMS films and the TENG composed of high dielectric constant PFS (8%) composite films delivered an enhancement of open circuit voltage and the short-circuit current by 4.57 and 5 times. On the basis, the nano bumps of 10–30 nm are created on the surface of the films by surface etching, the bumps can increase the contact area of TENGs, resulting in a maximum U_{oc} of 910 V, I_{sc} of 25 μA and maximum power of 0.25 mW/cm^2. The investigation of the surface structure of the PFS composite films by AFM has revealed that the synergistic combination of multiple nanoscales (10–30 nm, 60–80 nm, 300–500 nm) is more favorable to achieve high performance of TENGs. The Gaussian fit showed that the optimum output performance of TENG was achieved in the roughness range of 10–30 nm in the PFS composite films system. Finally, we also shown the application of wearable devices based on PFS (60 s)-TENG for powering various commercial electronic devices, which can effectively collect human mechanical energy and stably charge a variety of capacitors. What’s more, the surface structure composed of multiple nanoscales endows PFS with excellent superhydrophobicity. The PFS films remained fine performance after 40,000 drops of water and 90 days. This work achieves high performance and water resistance of TENG, demonstrating great potential in the field of self-powered waterproof wearable electronic devices.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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